

PUSHPINDER SEKHON

 <https://pushpindersekhon10.github.io/portfolio-website/>

Education

University of Western Ontario

2022 - 2027

Bachelor of Engineering Science in Software Engineering - Deans Honor List 2022-2025 - 86% - Cum Avg London, Ontario

Experience

Ontario Power Generation

May 2025 - Present

Engineering Intern

Oshawa, Ontario

- Conducted in-depth reviews of **DCC real-time operating system** code for **nuclear plant control systems**, verifying compliance with **hard real-time requirements** and reinforcing system reliability to understand reactor safety systems
- Designed and implemented a **probabilistic safety assessment** tool for predictive and planned maintenance-related risk analysis and out-of-service items, and developed a **deterministic scheduler** to analyze upcoming week's maintenance workloads, accelerating engineering workflows by **50 %** and saving **\$400 per week** in labor costs

Royal Bank of Canada

September 2024 - December 2024

Design Process Intern

London, Ontario

- Participated in RBC led collaboration with Western Engineering to prepare solutions for **indigenous water issues**
- Learned **design thinking** principles presented by RBC to develop critical thinking and problem solving skills
- Connected with RBC technical professionals in order to learn about **cybersecurity** threats facing the banking industry

Western Engineering Concrete Canoe Association

Aug 2024 - Present

Software Developer

London, Ontario

- Engineered an **image processing** module using ImageJ, applying **vector distance algorithms** derived from **calculus-based Euclidean norms** to analyze mesh structures and accurately match regions to a user-selected color
- Developing application which increases **calculation efficiency** of engineers by **70 %** compared to past system

Projects

Real Time Messaging Application | C++, JavaScript

[Github Link](#)

- Architected **high-throughput, multithreaded** WebSocket server in C++ using Boost.Asio/Beast, mastering non-blocking **socket I/O**, event-driven reactor patterns, and **thread synchronization** for low-latency communication
- Engineered **concurrency control** for SQLite message persistence by designing **custom file-lock detection** with retry/backoff logic, demonstrating expertise in OS-level locks, **race condition prevention**, and thread coordination
- Integrated secure session management over raw **TCP** sockets—including **SHA-256 hashing** and per-connection state handling—showcasing proficiency in **POSIX socket APIs**, and **OS-level security** practices

Kernel-Style Peer Routing & Heartbeat Scheduler | JavaScript

[Github Link](#)

- Architected **event-driven** routing subsystem in Node.js with 16-bit **Blake2s256-hashed** peer IDs and XOR-distance k-buckets, mirroring **OS kernel routing table designs** to achieve **O(1)** bucket insertion and efficient lookup
- Implemented lightweight **binary IPC protocol** with manual buffer framing, header parsing, and embedded JSON metadata, akin to designing a **custom kernel message-passing** interface for peer handshake and synchronization
- Built a **watchdog-style** heartbeat scheduler using precise timer interrupts and TCP sockets to monitor **peer liveness**, validated system reliability via automated test harness, in similar way to **kernel regression testing**

Kernel-Level TCP/UDP Behavior Testbed | Mininet, VirtualBox, NetEm, Wireshark, Python

- Used VirtualBox and configured bandwidth and latency constraints to emulate **kernel-level network conditions**
- Evaluated **UDP's fire-and-forget** behavior versus **TCP's reliable, bidirectional handshake** under **10 %** packet loss by instrumenting Netem, capturing packets with Wireshark, and analyzing retransmission and window-flow mechanics.
- Developed a Python-driven **analytics pipeline** to parse Wireshark TCP traces and **visualize CUBIC congestion** window dynamics, yielding actionable insights for tuning the OS's **TCP congestion-avoidance algorithm**

Technical Skills

Languages : Java, Python, C++, C#, JavaScript, Visual Basic (MS office), MATLAB

Developer Tools: VS Code, AWS EC2, IntelliJ, Unity, Jupyter Notebooks, Mininet, Wireshark, Packet Tracer, Linux

Interests

Ice Hockey, Canadian Geography, Lord of the Rings, Computer Games, Classic Films, Medieval and Ancient History, Linux